Angled launches

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Use components to analyze an angled projectile.

SUCCESS CRITERIA - I can...

1. Separate the initial velocity into two components.
2. Use vertical motion to determine flight time.
3. Explain the assumptions behind range comparisons.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Warm-up: bring back an earlier idea

A downward-velocity fit goes from 0.3 m/s at 0.01 s to 3.24 m/s at 0.31 s. Find g and explain why fitting an interval can be better than one noisy point.

Marzano check - circle: ① ② ③

02 Worked example: explain the model

Reference: $v_{x,\,i} = v_i \cos\theta$; $v_{y,\,i} = v_i \sin\theta$; $t_{flight} = 2\,v_{y,\,i}/g$
range = $v_{x,\,i}\,t_{flight}$.
θ above horizontal; g = 10 m/s². Equal launch/landing height; no air resistance.

Worked reasoning: A projectile has initial components 4 m/s horizontal and 10 m/s upward. With g=10 m/s² it reaches the top in 1 s and returns to launch height in 2 s; range=8 m.

Your turn: Sketch its path and velocity arrows at launch, top and return.



Add velocity / acceleration arrows.

Marzano check - circle: ① ② ③

03 Read and interpret the model

At the highest point, which velocity component is zero and which remains? Is acceleration zero?



Add velocity / acceleration arrows.

Marzano check - circle: ① ② ③

04 Apply the physics

A ball launches from level ground at 20 m/s, 30° above horizontal, and lands at the same height. Use g = 10 m/s² and ignore air resistance. Draw the initial velocity components. Find flight time, range and maximum height.



Add velocity / acceleration arrows.

Marzano check - circle: ① ② ③

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Continue down each column.

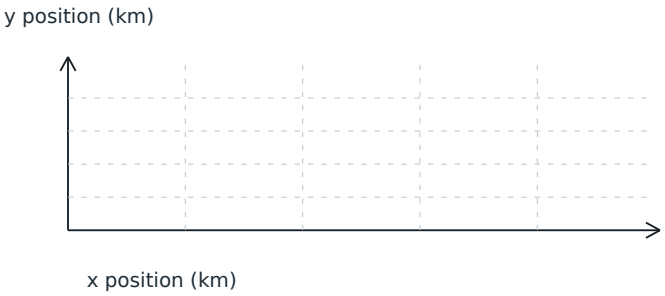
05 Explain your reasoning

Why is doubling time to the top valid only when the projectile returns to launch height in this ideal model?

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

A simplified model predicts position (-70, 90) km with bounds ±7 km in x and ±9 km in y. Write a short briefing giving both intervals and two reasons it is not a real asteroid collision forecast.
Label the given and predicted quantities on the model.



Marzano check - circle: 1 2 3

07 Exit check

For speed 20 m/s, compare ranges at 30° and 60° on level ground. Explain which has longer flight time. Label the model with the known quantities before calculating. Ignore air resistance; use g = 10 m/s² when needed.

Separate horizontal and vertical motion

Label x and y; state positive directions.



Velocity:

Acceleration / force:

Marzano check - circle: 1 2 3

08 STRETCH: test the model

For a launch at 20 m/s from level ground, find the two angles giving an ideal range of 20 m with g=10 m/s². Compare flight times. Label the model with the known quantities before calculating. Ignore air resistance; use g = 10 m/s² when needed.



Add velocity / acceleration arrows.

Marzano check - circle: 1 2 3